

Ben Eze

Email: be121@ic.ac.uk

Website: <https://ben-eze.github.io/>

LinkedIn: <https://www.linkedin.com/in/ben-eze-5a41661ba>

GitHub: <https://github.com/Ben-Eze/>

Aeronautical Engineering graduate from Imperial College London, currently working at the robotics startup *Rogue Cortex* and starting a PhD at Imperial in scientific machine learning for weather forecasting in December 2026. Research interests include physics-inspired model architectures and methods for interpreting their internal mechanisms to improve robustness.

Research Experience

Master's Thesis – Imperial

Oct 2025 – Jun 2026

Exploring methods for flow field reconstruction from sparse measurement windows. Presented my initial findings in a paper, accepted to ICCS (International Conference on Computational Science) 2026 and Springer LNCS Series.

Undergraduate Research Opportunities Programme (UROP) – Imperial

Jul – Sep 2024

Developed a high-order meshing application in C++ to interface between the commercial software ANSA and Imperial's in-house tool 'NekMesh', for use in turbomachinery simulations. Presented to engineers at Rolls Royce and ANSA, ensuring solutions satisfied all parties.

Additional Personal Projects

- Architected and trained a character-level transformer model from scratch (in PyTorch) achieving a validation cross-entropy loss of 1.4 on the TinyShakespeare dataset.
- Developed an interactive 2D 'finite difference' fluid simulation in Python, achieving >125 FPS on a 128×128 grid by leveraging NumPy vectorisation and JIT compilation.

Work Experience

Aerospace Engineer Intern – Rogue Cortex

Jun 2026 – Present

Building and testing a *developer drone* to be released in Autumn 2026, designing everything from the structural components and wiring to the testing rigs and software.

Aerospace Engineer Intern – BAE Systems

Jun – Sep 2025

Modelled the electronic warfare capabilities of various drones in STK (systems toolkit) to inform mission strategy in projects within the BAE-Lockheed Martin partnership.

Maths and Physics Tutor – Self-employed

2021 – Present

Self-employed tutor with 350+ hours of experience, with tutees from primary school to college level.

Education

MEng Aeronautical Engineering – Imperial College

2022 – 2026

- First Class Honours, 83% overall, ranked 3rd in the cohort
 - Year 1: 79% (ranked 5th), Year 2: 85% (ranked 2nd), Year 3: 75%, Year 4: 90% (ranked 1st)
 - Awarded the *Project Merit Prize (Computational)* within the department awarded for the computational Master's project of the highest merit (with a grade of 95%)

City of London School

2013 – 2021

- A-levels: $3 \times A^*$ AS-level: $1 \times A$ GCSEs: $11 \times A^*$

Key Skills

Collaboration

Learned how to ensure maintainable codebases on large-scale projects using Git through a research project at Imperial and an internship at Nationwide Building Society. Applied these practices to university group projects to enable efficient and reliable collaboration.

Communication

Have honed the ability to simplify complex concepts clearly through for years of experience tutoring mathematics. Consistently enable students to achieve grades above their initial targets.

Leadership

Coordinated committee decision-making, competition logistics, and budgeting as President of the Imperial Ice Hockey Society. Improved organisational structure led to a 28% increase in membership over the previous record, while tighter financial oversight eliminated £3,500 of club debt. Organised a 600-spectator Varsity match, with Imperial ticket sales up by 35% from the previous year.

Technical Skills

Python (NumPy, PyTorch, TensorFlow, SciPy), C++, Arduino, Git, CFD (Star-CCM+), numerical methods for PDEs, CAD (Fusion).

Extra-curricular Activities

London Defense Tech Hackathon: 1st place

2024

- Developed a concept design for countering radio frequency jamming systems for use in the war in Ukraine, a tool small enough to be attached to a drone.

President/Secretary/Social Secretary – Imperial College Ice Hockey

2023 – 2026

- As president, oversaw a 60-member society with two teams competing nationally across Great Britain.